How-To Library

Informatica

# Using Address Verification in QuickCapture Mode

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# Abstract

This document provides instructions and guidance on how to integrate Address Verification QuickCapture functionality in a user interface.

# Supported Versions

- Informatica Address Verification (On-Premises) 6.1 and later

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# Introduction

You can use the QuickCapture process mode in Informatica Address Verification to receive address suggestions in real time as you enter an address. This How-to Library article provides you with information on how to configure an input address for QuickCapture analysis and also provides information on how you can implement QuickCapture mode in your application environment.

Address Verification provides status codes and other properties, such as ResultGroup and AddressCount, that can help you better understand your results. Additional properties, such as MaxResultCount, can impact your results. Read this article to learn more about the properties.

# QuickCapture Setup

Use the QuickCapture mode in Address Verification to receive address suggestions as you enter an address.

The MaxResultCount property controls the number of results that Address Verification returns in the output. Set the MaxResultCount property to an appropriate value to receive the address suggestions in QuickCapture. For more information about MaxResultCount property, see "MaxResultCount" on page 3.

If you are an Address Verification (On-Premises) user, ensure that you fully preload the QuickCapture (QCP) reference data files for maximum responsiveness and user experience.

# When to call the service or the API

As best practice, do not call the service after every keystroke. With minimal input, you are unlikely to receive useful results. Decide on the number of input characters that will trigger a call from your application to the service. For example, you might permit the user to enter at least three characters before calling the service.

Allow for a latency period when calling the service, for example 100 ms to 200 ms. If the user enters multiple characters within the latency period, you do not need to call the service frequently.

# QuickCapture Input

Use the SingleAddressLine element and the Country element to enter an address in QuickCapture mode.

Consider the following rules and guidelines when you enter an address in QuickCapture mode:

- Country is a mandatory input element in addition to SingleAddressLine in QuickCapture mode. You can find the Country element under **IO** > **AddressElements** and the SingleAddressLine property under **IO** > **PreformattedData** in the AVJob.schema.json file.
- Address Verification ignores the SingleAddressLineDataType property in QuickCapture mode.
- Optionally, you can submit geocoordinates or an IP address as a physical location for the address. Submit an IP address in IPv4 or IPv6 format.
- Do not enable address enrichments during the QuickCapture process, as they increase response times. You can set address verification properties such as Casing and Preferred Script to meet the specific requirements of your use case.

## Input Order in QuickCapture Mode

The input order for data values in QuickCapture mode is flexible. You can enter the addresses in bottom-up or top-down manner, using the country-specific address formats or the address format of your country.

While some random input combinations may work, data values from different types of address information must not overlap in the input address. For example, an input of `HNO POSTCODE STATE STREET LOCALITY` cannot deliver good results, because the house number and street information are not adjacent to each other.

# Important Properties

Pay attention to the properties below when you configure the address verification engine for quick capture processing.

## *MaxResultCount*

The MaxResultCount property controls the number of suggestions that Address Verification can return for a single address. Find the property under **Parameters** > **CountrySets** > **Result** in the `AVJob.schema.json` file.

The default number is 20, which is typically enough for a desktop application. On a mobile application, you may decide to set the property value to no more than 10, as mobile users may not want to scroll through many results. You can set the property to return up to 100 suggestions for an input address. Do not use set the property to a value lower than 5.

You will receive fewer address suggestions when there are fewer matches in the reference data, but in most cases the number of result is capped at the number that you configure in the property.

## *QuickCaptureResultFilter*

QuickCapture mode allows application users to select a non-unique address suggestion and obtain the list of addresses that Address Verification identifies. This drill-down process can continue through multiple levels of address suggestions.

To enable the drill-down capability, add the `QuickCaptureResultFilter` property to your engine integration. Find the property under **Parameters** > **Country Sets** > **Result**.

The `QuickCaptureResultFilter` property is an array of integers where every integer represents an index to a non-unique address suggestion. Configure your software application to read the integers on the property. When index values are present in this array, the engine discards all other address suggestions and expands the suggestion that the user selects.

**Note:** If you enter additional data while clicking through the drill-down suggestions, the indexes will not match the results for the latest input. In this case, the application must clear the `QuickCaptureResultFilter` array values and restart the click-through process with the latest address results.

## *NumericRangeExpansion*

The NumericRangeExpansion element specifies how Informatica Address Verification returns address suggestions when the input address data matches a range of addresses in the same building or street. For example, Address Verification might determine that an input address matches a range of house numbers on a given street. In such a case, Address Verification can return the upper and lower numbers in the available range in a single suggestion.

Find the NumericRangeExpansion element under **Parameters** > **CountrySets** > **Result** in the `AVJob.schema.json` file.

Address Verification can apply numeric range expansion to house number data, building data, and sub-building data. Address Verification can perform range expansion in QuickCapture mode and Interactive mode.

### NumericRangeExpansion Properties

To define how Address Verification handles number ranges, configure the following properties on the NumericRangeExpansion element:

**RangesToExpand**

Specifies whether to expand number ranges in address suggestions from countries for which individual house number, building, or sub-building values are available in the reference data.

**RangeExpansionType**

Specifies the scope of range expansion in suggestion lists.

### RangesToExpand

The RangesToExpand property has the following options:

**All**

Expands the house number, building, or sub-building values to match the individual house numbers available in the reference data.

**OnlyValid**

Expands the ranges only when the reference data contains the complete list of valid numbers in a range.

**None**

Does not expand any range. The default value is None.

When the complete list of house numbers is not available in the reference address database, OnlyValid works the same way as None.

### RangeExpansionType

The RangeExpansionType property has the following options:

**Full**

Expands all the items in the suggestion list, depending on the current RangesToExpand options.

**Flexible**

Expands the initial items in the suggestion list, depending on the current RangesToExpand options.

If you set the RangeExpansionType property to Flexible, the internal engine logic decides on the number of results to expand and how many to keep as ranges without exceeding the maximum result count value. Therefore, the suggestion list might contain both expanded and unexpanded ranges if you set the RangeExpansionType property to Flexible.

## Numerical Range Formats

Address Verification uses two periods (..) to indicate a numerical range in QuickCapture mode. Address Verification uses a dash symbol (-) to indicate a numerical range in interactive mode.

For example, 15500..15508 can indicate that houses from 15500 through 15508 on a given street are candidate matches for an input address in QuickCapture mode.

When the Address Verification engine returns multiple ranges for an input address in a single suggestion, each range is separated by a semicolon. For example:

23..27;30..36

The engine can return multiple ranges in a single suggestion.

If the engine reaches the MaxResultCount limit for a given address and additional ranges remain in the reference data, the engine adds three periods (...) after the values in the final suggestion. This indicates that additional address suggestions are available in the reference data.

**Note:** Generally, Address Verification returns even and odd numbers in a single range. However, Address Verification might return even-numbered and odd-numbered ranges in separate records for a single input address. For example, 1..9 MAIN STREET and 2..10 MAIN STREET may indicate odd-numbered and even-numbered houses respectively.

## Rules and Guidelines for Numerical Ranges

Consider the following rules and guidelines for expanding numerical ranges:

- Address Verification ignores the RangeExpansionType property when the RangesToExpand property value is None.
- In jobs that run in QuickCapture mode, Address Verification uses All as the RangesToExpand property value and uses Flexible as the RangeExpansionType property value. Address Verification ignores the physical properties of the NumericRangeExpansion element in QuickCapture mode.
- Each address suggestion that the engine returns can identify a single address, a range of addresses, or multiple ranges. The combination of single addresses and ranges in the suggestion list depends on the RangesToExpand and RangeExpansionType properties and on the MaxResultCount property value.
  The engine may return single addresses in the initial address suggestions and return addresses in ranges in subsequent suggestions.

# *PreformattedData*

Address Verification includes a set of semi-fielded and unfielded output formats that are suitable for the addresses that QuickCapture mode generates. You can select one or more of the formats for your address job. You can view the

default values for the preformatted data outputs in the AVJob.schema.json file of an on-premises installation. Find the properties under **CountrySets** > **OutputDetail** > **PreformattedData**.

The following code fragment shows the default settings on the PreformattedData object:

```
"PreformattedData": {
   "type": "object",
   "default": {},
   "properties": {
      "PostalFormattedAddressLines": { "type": "boolean", "default": true },
      "PostalDeliveryAddressLines": { "type": "boolean", "default": true },
      "PostalRecipientLines": { "type": "boolean", "default": true },
      "PostalLocalityLine": { "type": "boolean", "default": true },
      "StreetWithNumber": { "type": "boolean", "default": false },
      "NumberAndSubBuilding": { "type": "boolean", "default": false },
      "SingleAddressLine": { "type": "boolean", "default": false },
      "SingleAddressLineDelimiter": { "type": "string", "default": "Semicolon" }
   }
},
```

Consider the following rules and guidelines when you configure the PreformattedData properties:

- The SingleAddressLine field does not include data that is not part of a formatted address.
- You can configure the SingleAddressLineDelimiter property on the PreformattedData element. The delimiter separates the address lines that Address Verification returns in SingleAddressLine output.
- You can also configure the properties to return address suggestions from individual elements. However, Informatica does not recommend this. The QuickCapture results are more legible in a single address line.

# QuickCapture Output

There are important differences between the output from QuickCapture mode and the output from other address verification modes. You must understand the differences to properly implement QuickCapture mode in a user interface.

## *Process Status Codes*

When you submit an address in QuickCapture mode, Address Verification returns the following values in the ProcessStatus field: P, F, or I. The status codes provide important information on the list of suggestions that the service returns. Unlike their operation in other modes, the status codes do not distinguish between good and bad addresses.

A status of P indicates that one or more of the suggestions returned is an incomplete address or an aggregated address. If the user selects an incomplete or partial address from the suggestion list, it is not the final step in the process. This is important for the user interface, because the user is expected to provide additional information after they select the address. Typically, the first responses return a P status.

A status of F indicates that all of the returned addresses are complete addresses. A complete address has an AddressCount value of 1.

A status of I indicates that Quickcapture mode returned no addresses.

When the user selects a complete address from the suggestion list, the quick capture process is complete from the user's point of view. You do not need to call the service again.

## *Result Groups*

Address Verification can sort address suggestions into groups based on the element in each suggestion that most closely matches the input data. Address Verification returns data in groups exclusively in QuickCapture mode.

Find the property in the `AVJob.schema.json` file under **IO** > **Outputs** > **Results** > **Variants** > **StatusValues**.

Result groups are useful because the initial input data in QuickCapture mode can match hundreds or even thousands of candidate addresses in the database. This can happen when minimum input information is presented. Additionally, an input value might match multiple elements. For example, if a user enters `Eisenhower` in a United States address, the input value might refer to a street, a building, or an organization. Instead of returning a list of address suggestions ranked only by the accuracy of the reference data match, Address Verification returns the list of suggestions in groups, displaying the best matches across the types of information that each group represents. For example, the results may be grouped by the organizations, buildings, and street names that include `Eisenhower`. When Address Verification finds more than one element that matches the input data, it gives precedence to the most local element.

You can also use result groups to distinguish between different sets of results in the user interface. A user interface designer might consider matches in certain categories to be more important than others. The designer can use the result groups to prioritize the results before displaying the data to the end user. However, Address Verification always returns the group with the best matches first. For example, if there is a 100% match in the Street group and 95% matches in the Building group, Address Verification returns the Street group first to make sure that the best match is at the top of the list.

While Address Verification returns the group with the best matches first, it otherwise returns groups in the following order by default:

- Organization
- Building
- Street
- DeliveryService
- PostalCode
- Locality
- AdministrativeDivision

You might decide not to return suggestions from a given group. For example, if a consumer is entering their own address, grouping address suggestions by organization might not be useful.

Depending on the suggestions identified in the database and on the MaxResultCount limit, Address Verification may not populate all result groups.

### Sorting Criteria in QuickCapture Mode

Within each group, QuickCapture mode considers the following factors when it sorts and returns address suggestions:

- The similarity between the input address and the reference address, as indicated by the MatchPercentage value.
- The distance between the input address and the reference address, if the distance can be calculated.

**Note:** For the purposes of sorting address suggestions in QuickCapture mode, the location of the reference address is estimated at the locality or post code level. The location is a very general measurement and is not mapped to the street or building level. This enables Address Verification to sort the addresses for example by city or town. In QuickCapture mode, address suggestions within the same city or town are effectively assigned the same distance from the input address.

If you provide an IP address or geocoordinates in the input address, the distance between the input geocoordinates and the geocoordinates of the reference address becomes the primary sorting criterion within each result group. As above, this location is estimated at the locality or post code level and is a very general measurement of the address location.

## Complete and Incomplete Addresses

An address suggestion that QuickCapture mode returns can be a single, complete address or an incomplete address that represents multiple complete addresses.

A suggestion that indicates multiple addresses is typically a street with multiple building options or a building with multiple sub-building options. For example, if the reference data contains 12 sub-buildings for a given house number, Address Verification returns a value of 12 in the AddressCount property.

The AddressCount property can also return the possible number of houses, buildings, or sub-buildings for output addresses that contain range-based house numbers, sub-buildings, or buildings.

The AddressCount property indicates the number of complete addresses that an address suggestion can represent. The number can be displayed in the user interface. It also indicates whether the capture process is finished.

An AddressCount value of 1 indicates that the address suggestion is a single, complete address. An AddressCount value greater than 1 indicates that the suggestion is not a complete address and that complete suggestions are available for the current suggestion. In this case, you can call the API again with the incomplete address suggestion as the input. Or, you may be able to expand onscreen the set of house number or sub-building suggestions so that the user can select an address.

For addresses in a Street, Locality, or PostalCode group, the AddressCount property returns an approximate value followed by a **+** symbol. The AddressCount property returns a **?** symbol when Address Verification has no information about the possible number of complete addresses, for example if the database contains no information about house numbers.

## Number Ranges

If an incomplete address suggestion represents a range of house numbers or sub-building numbers, QuickCapture mode can display the ranges in the suggestion. This enables the user experience designers to display suggestions in different ways. Address Verification uses two dots [..] to indicate a range of possible numbers.

An address suggestion can contain multiple ranges. If Address Verification cannot display all of the possible ranges for an address, it uses three dots [...] to indicate the presence of additional ranges in the reference data. For example, additional ranges may be present if the MaxResultCount property limits the number of suggestions that Address Verification returns for the address.

The following text shows the range format:

```
<Low Number 1>..<High Number 1>;<Low Number 2>..<High Number 2>;<Low Number 3>..<High Number 3>;...
```

The three dots that follow the third range indicate the presence of additional ranges in the reference data.

The following example illustrates sample ranges in the same format:

```
100..299;1000..1399;1700..1799;...
```

**Tip:** Do not display this information as-is in the user interface.

The information is intended for the user interface integrator, who can use it in the application in one of the following ways:

- The application may ignore the information that Address Verification returns for an address in the suggestion list and wait for the user to enter more information.
- The application may indicate that an address suggestion requires more information.

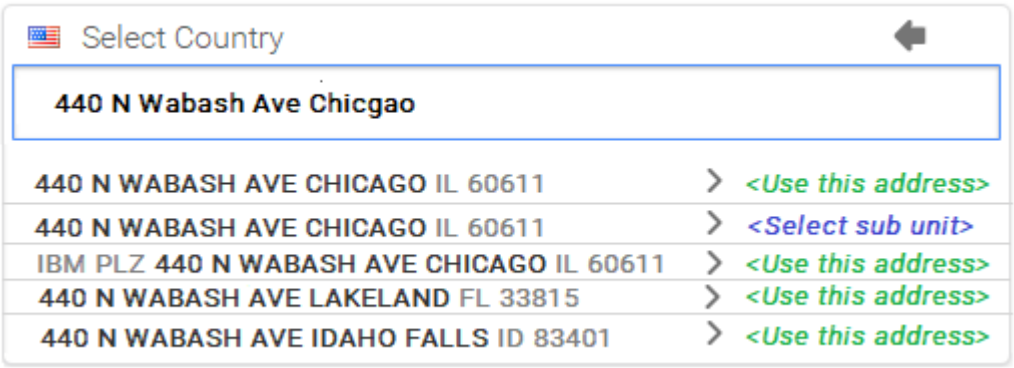
  When the user selects a suggestion, the application can do one of the following:

  - Display an address list for the user to select from (if the list contains fewer than 20 house number or sub-building options).

- Prompt the user to enter house or sub-building data.

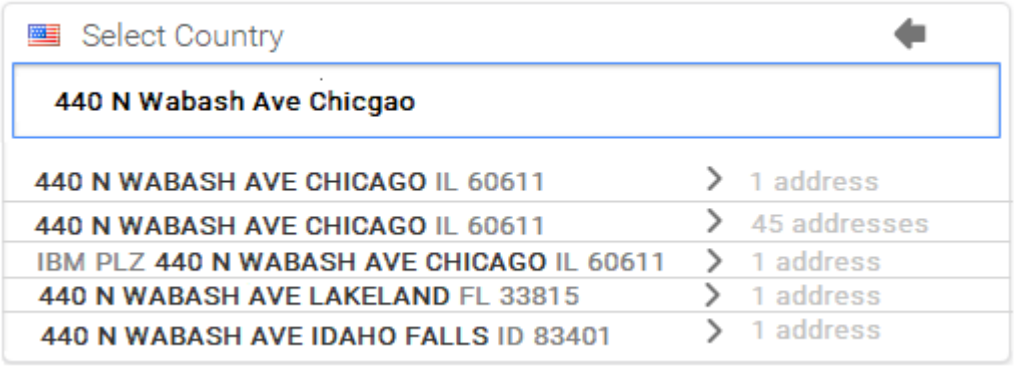
  For sub-building ranges, the integrator can usually retrieve the type of absent sub-building data for the address and customize a message around it, such as *Please enter a valid apartment number*.

  The following image shows an address suggestion that requires more address information:




- The application may display the number of addresses available for each address in the suggestion list.

  The following image shows the numbers of addresses available for the suggestions in a suggestion list:




**Note:** When an address suggestion specifies a set of possible numbers that are not in a range, Address Verification separates the numbers with a semicolon. If the AddressCount is 1, the address does not contain any ranges and is complete in its current state.

## *Sorting Results by Location*

When you include geocoordinates or an IP address in an input address, Address Verification considers the address location as a factor in sorting the address suggestions. Address Verification can sort the suggestions according to the distance between the geocoordinates of the input data and the geocoordinates stored for the locality or postcode of each suggestion.

If you submit an IP address with a partial address, Address Verification attempts to convert the IP address to geocoordinates and uses the geocoordinates to approximate the address location for the purpose of sorting the suggestions.

Bear in mind that an IP address may not identify a location with precision, and an identified town or district is not necessarily an indicator of the accuracy of a suggested address. This is particularly the case when you provide minimal address information as input or when the IP address is identified over a VPN.

# Work Flow

An address job might call the API multiple times during the process of capturing a user's address. One user might select a partial address from the list of address suggestions. Another user might continue to enter address information until they receive the results that they expect and only select an address at that final stage. It is important to understand if a 'click' in a suggestion list is the user's final step or a step in between.

If it is the final step, the process ends. You might call the API one last time with the selected address in batch, interactive, or certified mode, depending on your use case.

If the user selects an address result that is not yet a complete address, the suggested work flow is different. In such cases, the user likely expects to drill down into the address that they select. To facilitate this, call the QuickCapture service again with the address result that the user selected. You can also update the address in the input with the selected address.

Do not enable the enrichments during the quick capture process. Instead, enable the enrichments in the final call of the API to other processing modes.

## Example

The following example shows a valid address job request for a Germany address in QuickCapture mode:

```
{
"Login": "****",
"Password": "****",
"UseTransactions": "TEST",
"Request":
{
      "Parameters": { "Mode": "QuickCapture" ,
      "CountrySets": [
      {
        "OutputDetail": {
          "PreformattedData": {
            "PostalFormattedAddressLines": true,
            "SingleAddressLine": true,
            "SingleAddressLineDelimiter": "Space"
          },
            "SubItems": true
        },
        "Result": {
          "MaxResultCount": 20
        },
        "Standardizations": [
          {
            "Default": {
              "PreferredScript": {
                "Script": "Latin",
                "TransliterationType": "Default",
                "LimitLatinCharacters": "Latin1"
              },
              "FormatWithCountry": false,
              "CountryNameType": "NameEN",
              "CountryCodeType": "ISO3",
              "MaxItemLength": 255,
              "Casing": "PostalAdmin",
              "DescriptorLength": "Database",
              "AliasHandling": "PostalAdmin"
            }
          }
        ]
      }
    ]
  },
      "IO": {
         "Inputs": [
           {
              "AddressElements": {
                 "Country"     : "de"
```

```
                },
                "PreformattedData":{
                   "SingleAddressLine":"Röntgenstr"
                }
             }
          ]
         }
      }
   }
```

## *Work Flow Examples*

You can refer to the preceding example for the parameters to set for the addresses in the following examples. Bear in mind that you might receive different address results if you set any additional parameters for the addresses.

**United States address**

Consider the following address:

```
547 eisenhower
USA
```

The user receives a suggestion list with 20 results. Each address is complete, so the ProcessStatus value is F. If the user selects an address from the suggestion list, the process is finished.

Let's assume the user selects the following address from the suggestion list:

```
547 EISENHOWER ST DAVIS CA 95616
```

You can verify the address in certified mode and enable the enrichments. Address Verification returns the following address with geocoordinates in the output:

```
547 EISENHOWER ST
DAVIS CA 95616-3028
USA

Geocoordinates: 38.547582, -121.775144
```

**Germany address**

Consider the following address:

```
Roentgenstr.
DEU
```

The user receives a suggestion list with 20 incomplete results. Each address represents multiple house numbers, so when the user selects one of the results, they must further select a house number from the subsequent results.

Let's assume the user selects the following address from the suggestion list:

```
Röntgenstr. 2..4 97959 Assamstadt
```

You can expand the house number range "2..4" so that the user can select a complete address from the suggestion list:

```
Röntgenstr. 2 97959 Assamstadt
Röntgenstr. 3 97959 Assamstadt
Röntgenstr. 4 97959 Assamstadt
```

**Note:** A call to the API is not necessary. From version 6.1.0 onward, Address Verification automatically expands the address suggestions in case there are results left to populate.

## Iterative Quick Capture and the Cloud Interface

The QuickCapture process can be an iterative one. You may need to submit an address more than once before you find the address that you need. To facilitate the process, Address Verification can return a job token that allows you to submit multiple requests to the cloud interface on a single transaction charge.

Address Verification returns a job token in response to each new job request in QuickCapture mode. When you submit a job with the associated job token, you do not incur an additional charge.

A job token is an alphanumeric string that can contain up to 50 characters.

The following string is an example of a job token:

```
b89be474-318c-4da3-87c2-f15b2e43c2cb
```

You can resubmit a job with a given job token nine times in quick capture mode without additional cost. If you use the job token in further requests, or if the job token is not valid, Address Verification treats the request as a new transaction and applies a transaction charge accordingly.

## Author

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